

Yuan Si

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RESEARCH INTERESTS

AI for software engineering (AI4SE), LLM-based program repair, multimodal debugging, program analysis & verification for block-based programs, AI for computing education; analytic number theory, probability theory.

PUBLICATIONS

- **Yuan Si**, Daming Li, Hanyuan Shi, Jialu Zhang. *VisionScratch: LLM-Based Automated Feedback Generation using Code-Produced Videos for Scratch Programs*. Proceedings of the ACM on Software Engineering, Vol. 3 (FSE 2026), pp. 3534–3557. 
- **Yuan Si**, Simeng Han, Daming Li, Hanyuan Shi, Jialu Zhang. *ScratchEval: A Multimodal Evaluation Framework for LLMs in Block-Based Programming*. , ACM International Symposium on Software Testing and Analysis (ISSTA) 2026 (accepted).
- **Yuan Si**, Jialu Zhang. *ScratchLens: Lens-Parametric Behavioral Equivalence for Scratch Programs*. , 2026.
- **Yuan Si**, Ming Wang, Daming Li, Hanyuan Shi, Jialu Zhang. *EcoScratch: Cost-Effective Multimodal Repair for Scratch Using Execution Feedback*. , 2026.
- **Yuan Si**, Kyle Qi, Daming Li, Hanyuan Shi, Jialu Zhang. *Stitch: Step-by-step LLM Guided Tutoring for Scratch*. , 2025.
- **Yuan Si**. *A Microcanonical Phase Transition for the Collatz Affine Random Model* (2026). Submitted to *Forum of Mathematics, Sigma*. 
- **Yuan Si**. *Mixed Parity, Diagonal Denominator, and the Pell-Chord Genus-Five Obstruction for the Four-Corner Rational Distance Problem* (2026). Submitted to *J. Number Theory*.  
- **Yuan Si**. *Elliptic Decomposition of the Pell-Chord Genus-Five Obstruction for the Four-Corner Rational Distance Problem* (2026). 
- **Yuan Si**. *A Gentle Introduction to Optimization* (2024). Adopted as required reading in 3 university courses. 

EXPERIENCE

- **University Researcher - AI in Multidimensional Input** Remote, Waterloo, Canada
UWaterloo, ECE Department July 2025 – present
 - Led 9 first-authored papers on LLM-driven debugging, tutoring, evaluation, and formal verification for Scratch (VisionScratch published at FSE 2026, ScratchEval accepted at ISSTA 2026).
 - Built multimodal repair systems fusing gameplay video, execution feedback, and project JSON; developed the Stitch interactive tutor and the 100-project executable ScratchEval benchmark.
- **University Researcher - Game Theory & Graph Theory** On-site, Waterloo, Canada
University of Waterloo, Combinatorics & Optimization Department Dec. 2024 – March 2025
 - Derived Nash equilibria for multiplayer RPS via LP ; modeled Tesla charging placement via independent dominating sets ; analyzed Public Goods Games .
- **Microsoft Researcher - AI Trends** Remote, Vancouver, Canada
Microsoft - Microsoft Azure & AI, Research May 2024 – Aug. 2024
 - Investigated AI scribe technologies for healthcare and improved an insurance chatbot via systematic A/B analysis of user interaction data. 

EDUCATION

University of Waterloo Waterloo, Canada
HBM, Math / Data Science & Combinatorics Optimization & Pure Math (co-op) Sept. 2023 – April 2028

PROJECTS

- **ResearchOS** : Local-first research decision and execution system; turns a brief into runnable experiments, evidence-checked claims, and a packaged manuscript draft. FastAPI + React; dual-agent code worker; HITL approval gates. AGPL-3.0.
- **DevToolkit** : 43 zero-dependency single-file Python CLI tools across 9 categories (web, data, process, MCP, security, scaffolding).

SKILLS

- **Research**: Experiment design, human evaluation, ablation studies, benchmark construction, statistical analysis
- **AI/LLM**: Multimodal repair pipelines, LLM evaluation, prompt engineering, MCP tooling
- **Languages & Tools**: Python, JavaScript/TypeScript, C/C++, Java, Git, Linux, L^AT_EX, Magma, SageMath